SLTC REASEARCH UNIVERSITY

SEMESTER 05

NovaMart Retail Sales Dashboard

Power BI Practical Tutorial — Final Report

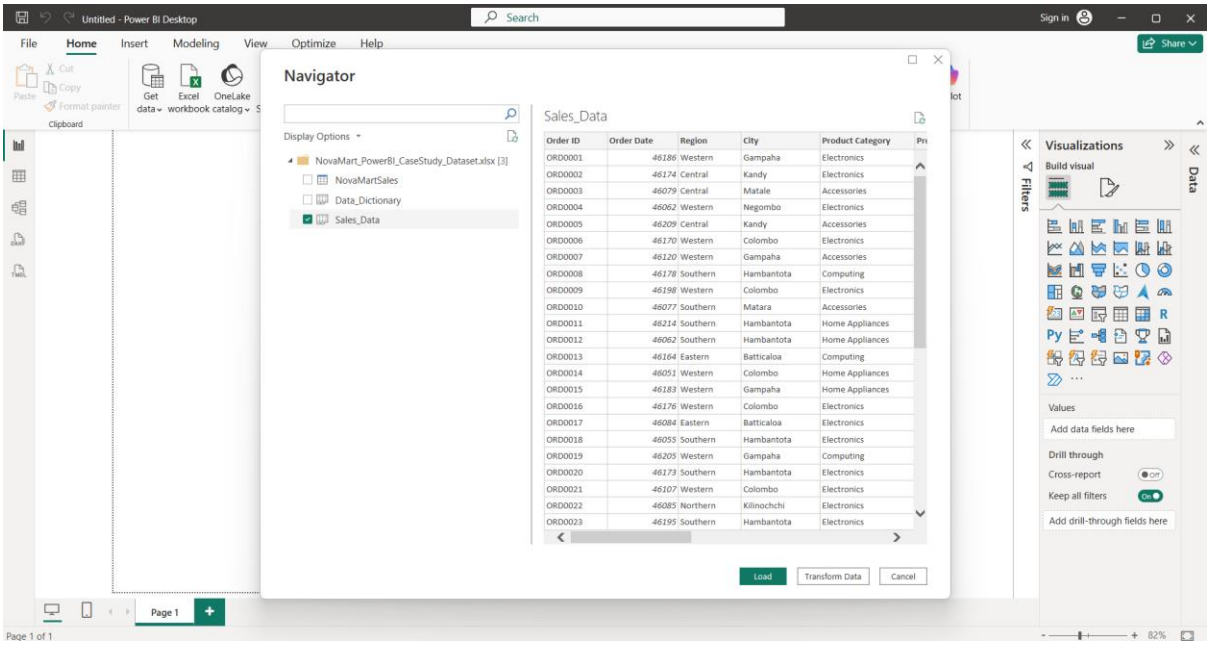
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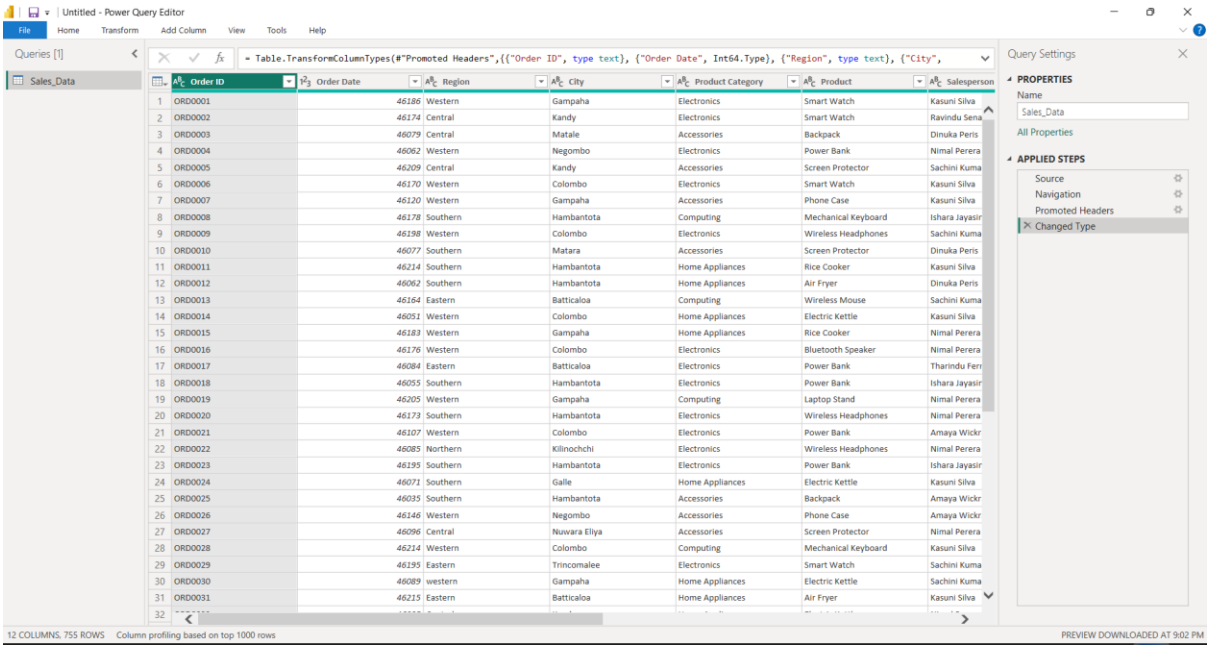
CIT308 - Data Visualization

Name: MM Fathima Sumla

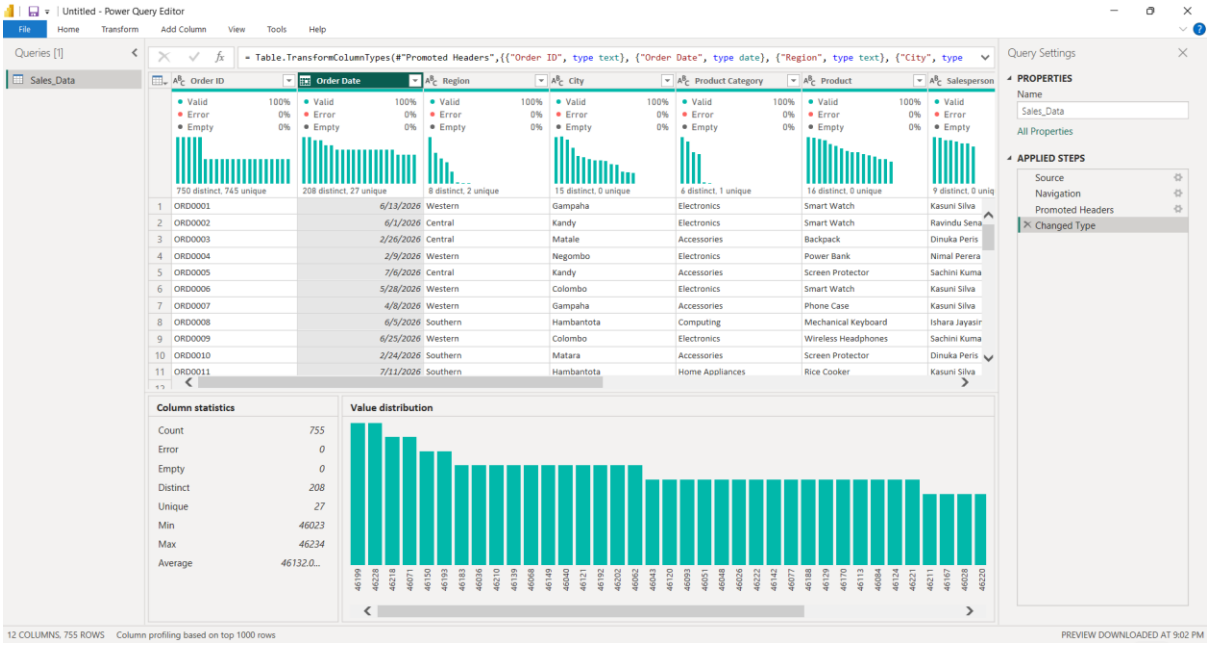
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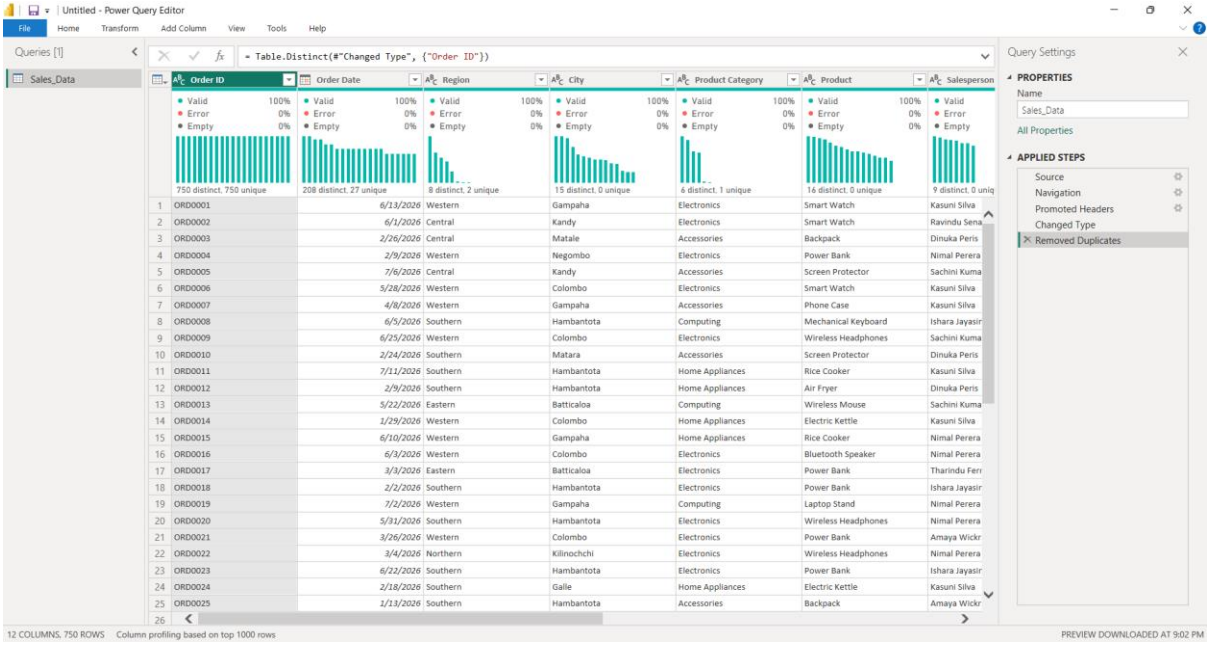
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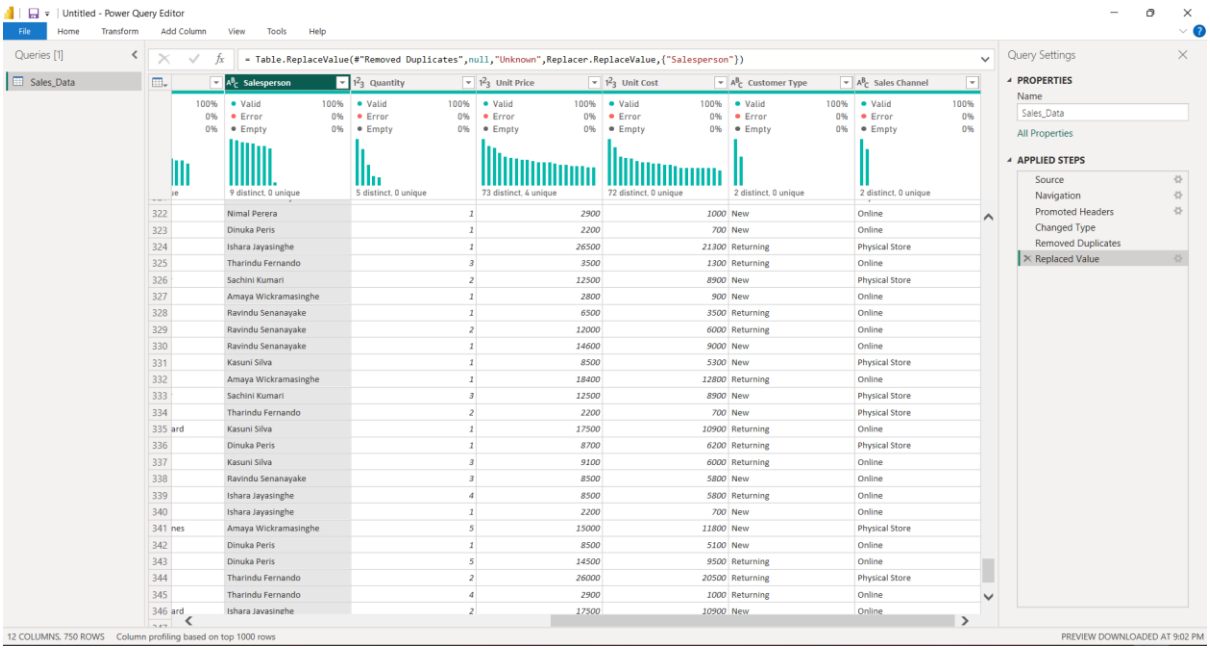


Step 2 - Inspect the Data

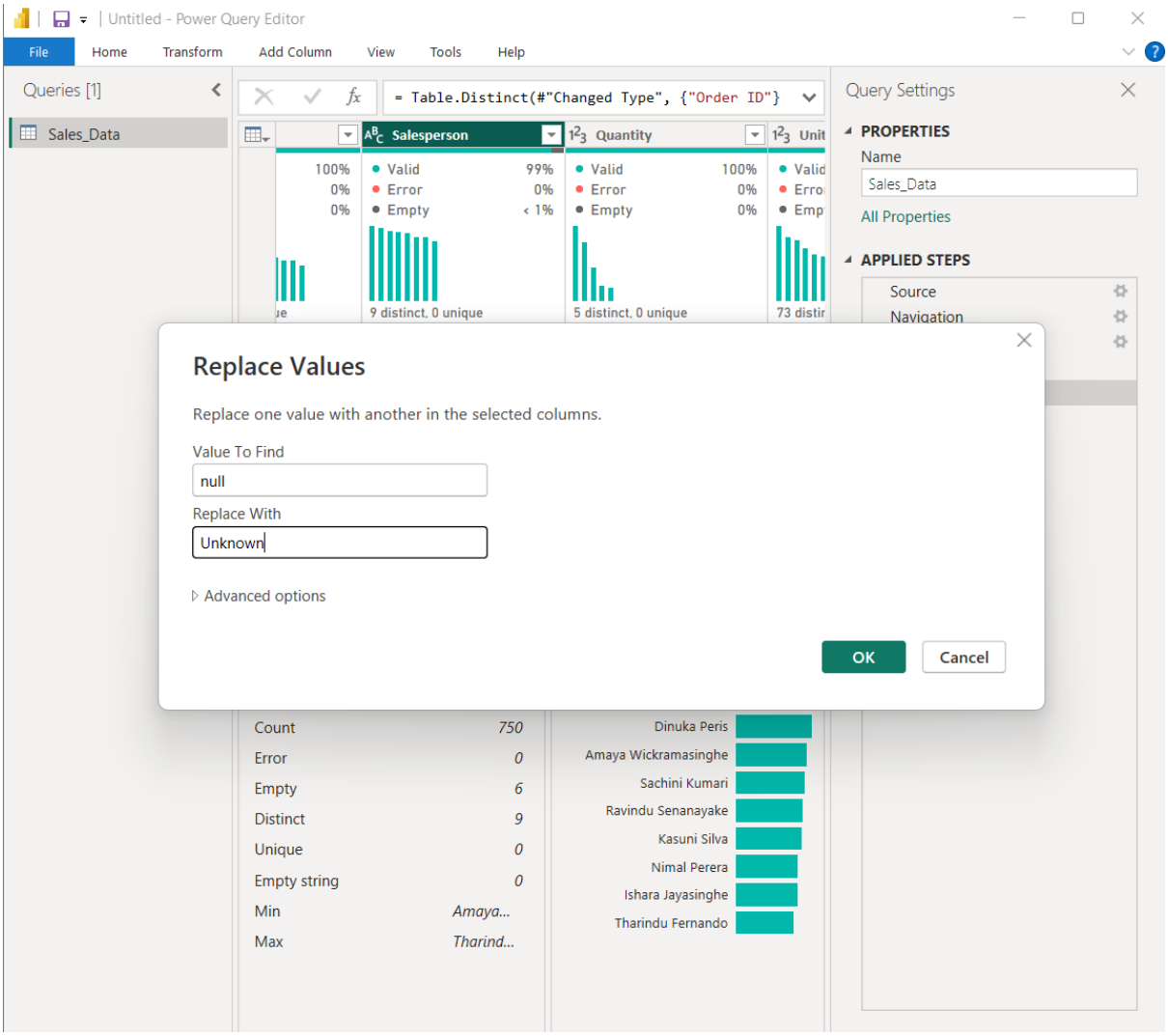


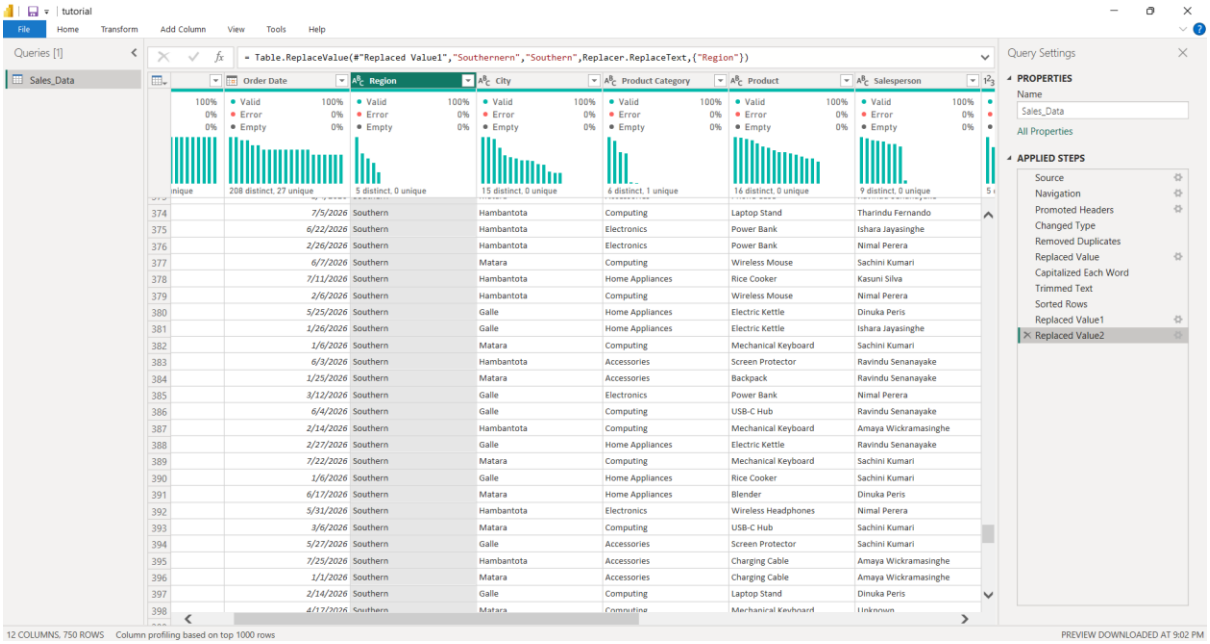


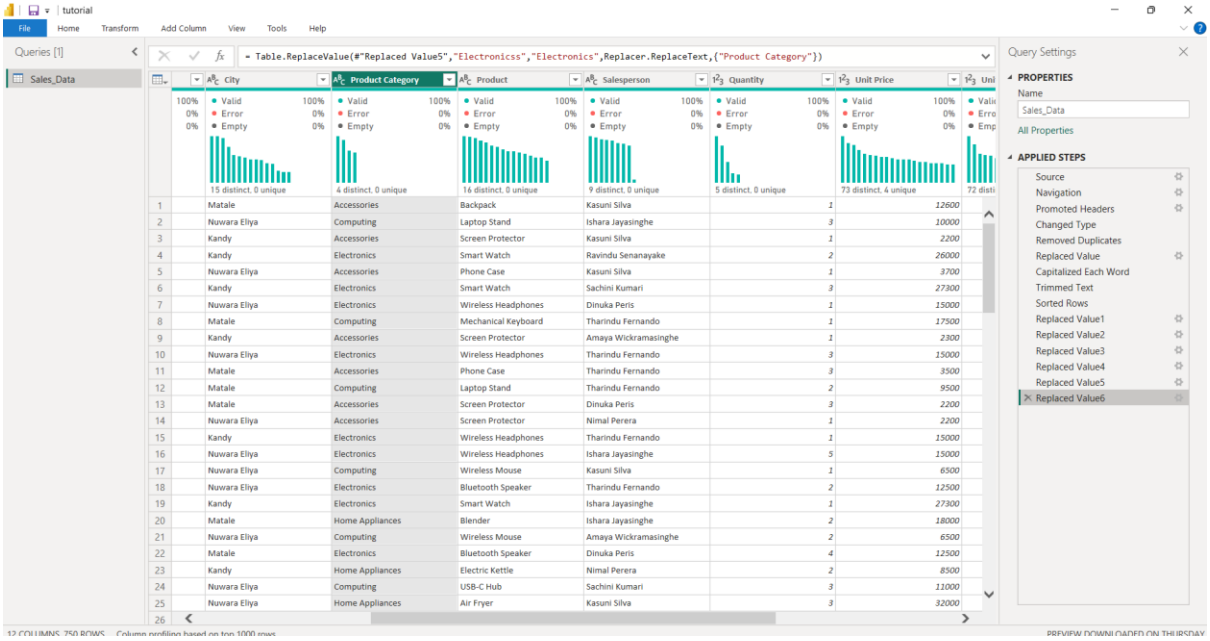
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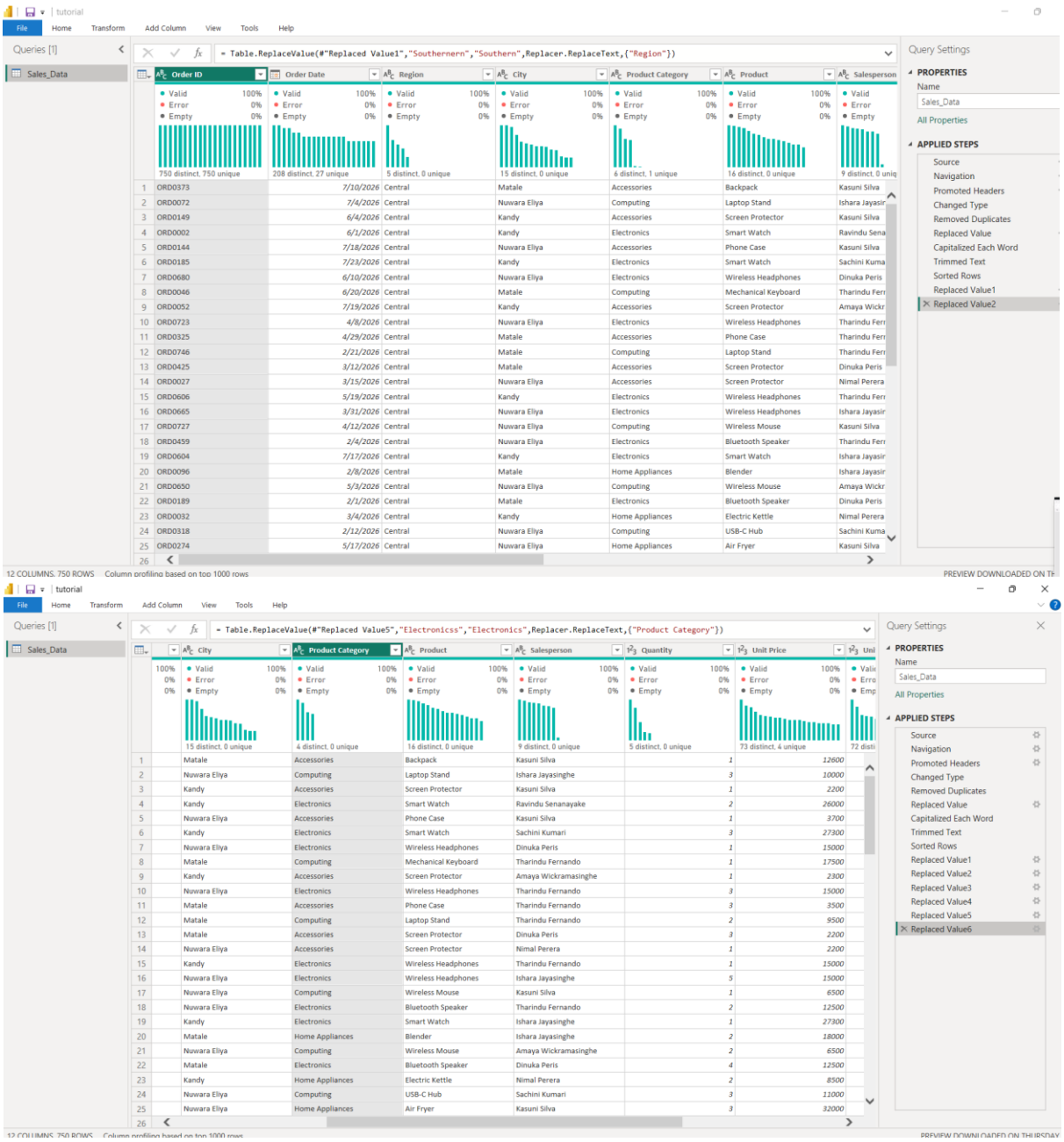


Step 4 - Clean the Dataset



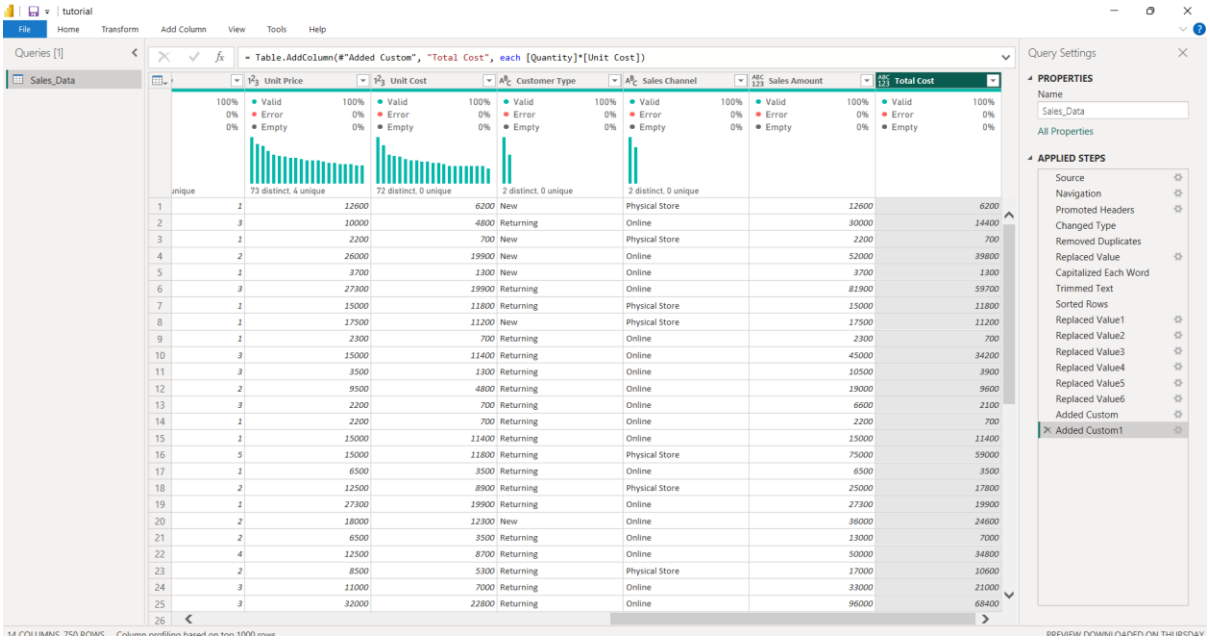

tutorial
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All Properties
APPLIED STEPS
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Promoted Headers
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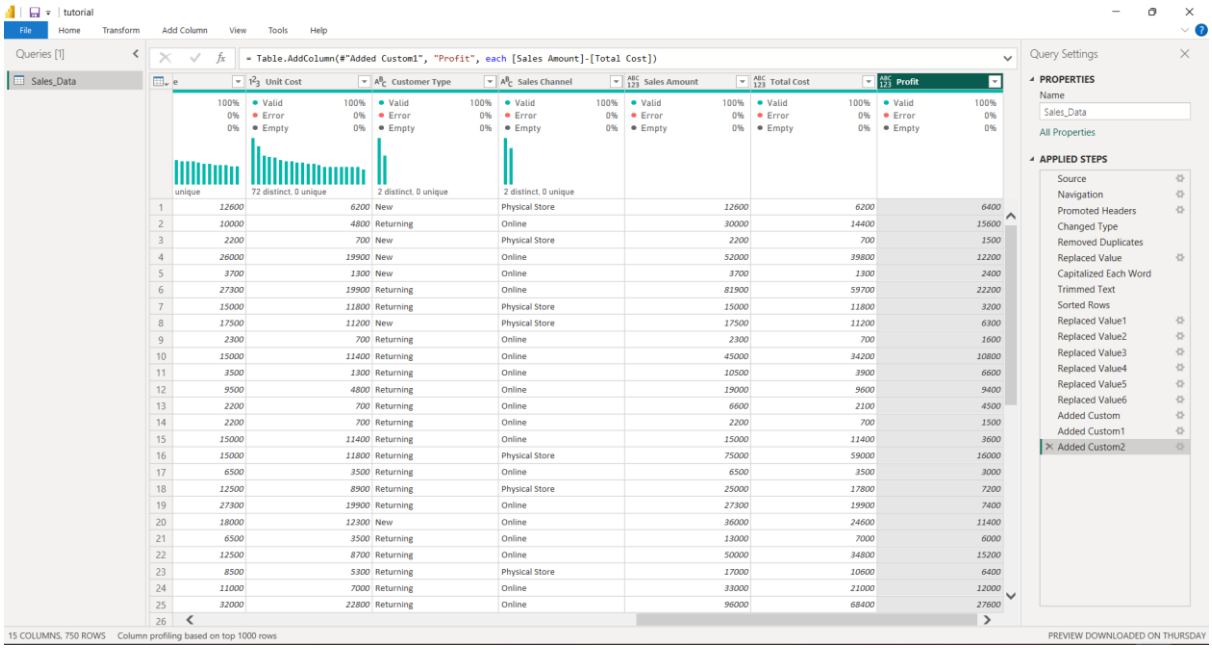

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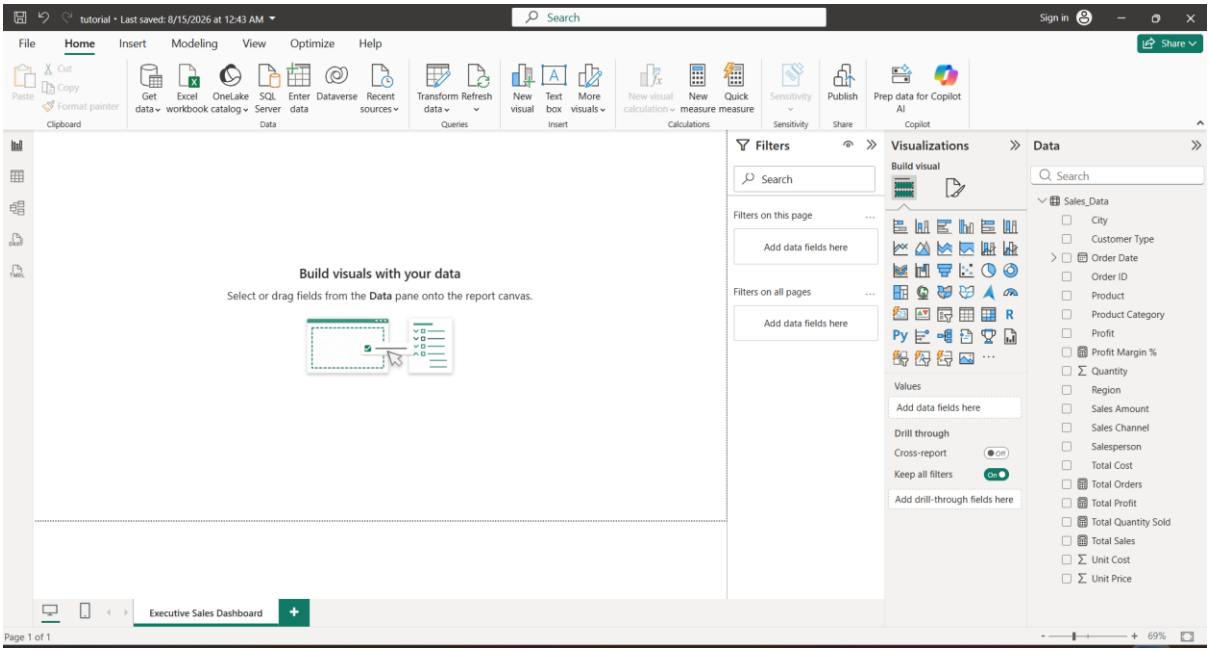

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12 COLUMNS, 750 ROWS
Column profiling based on top 1000 rows

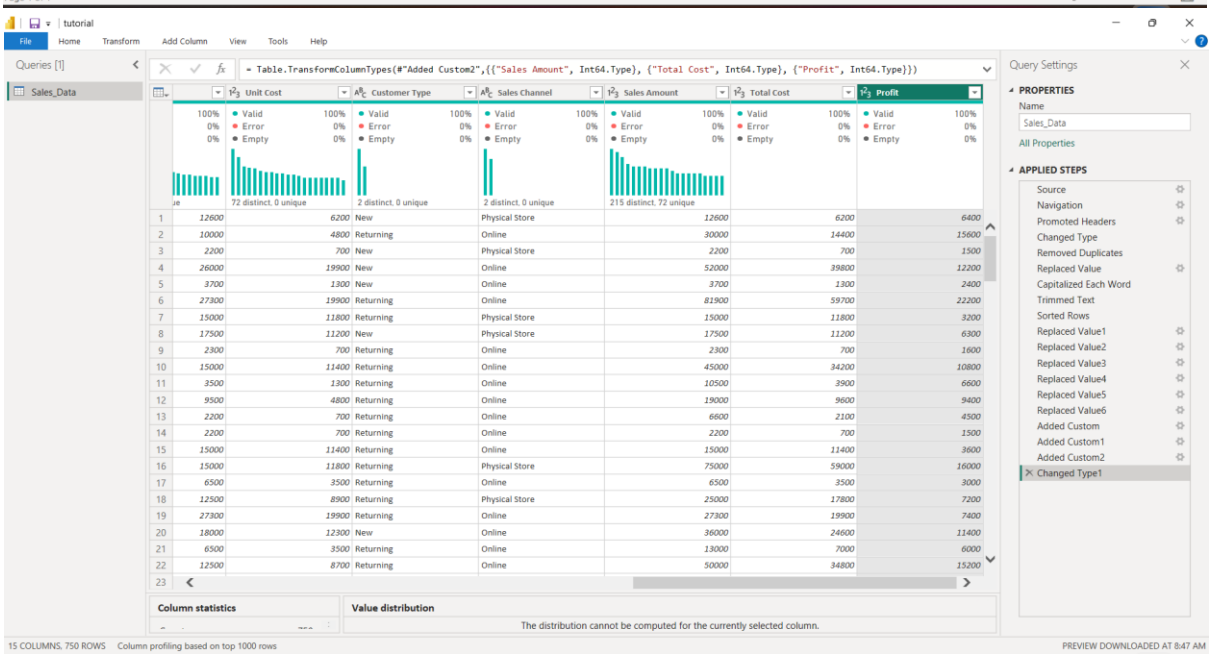
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Step 5 - Create Calculated Columns

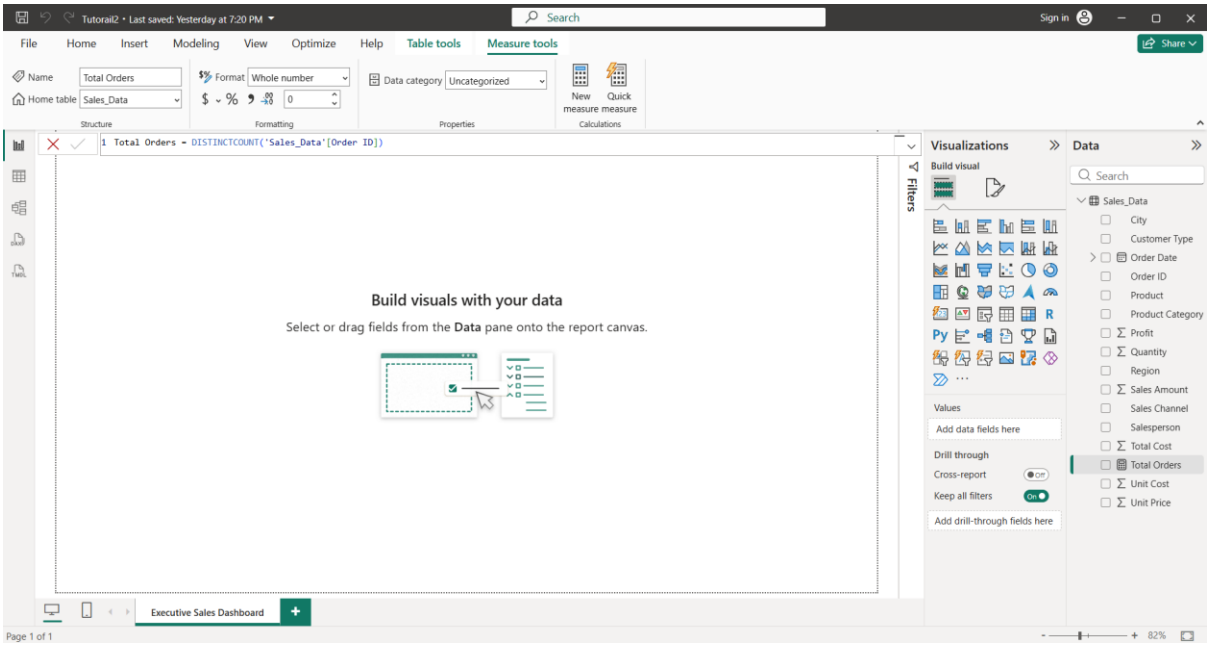


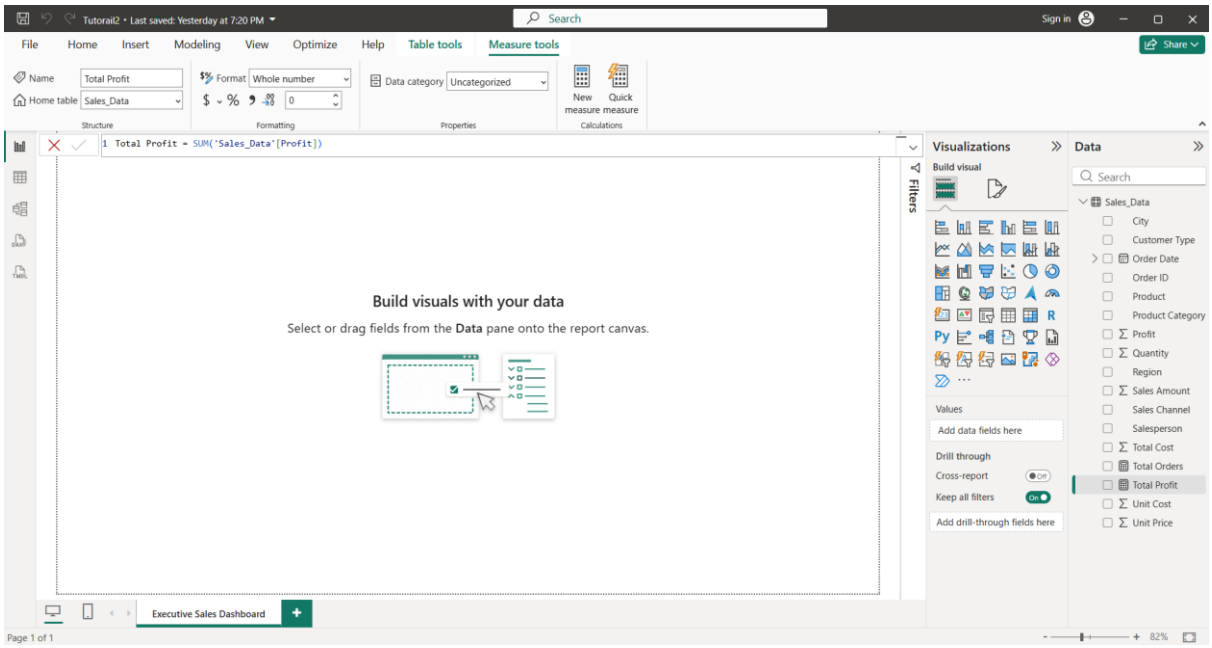


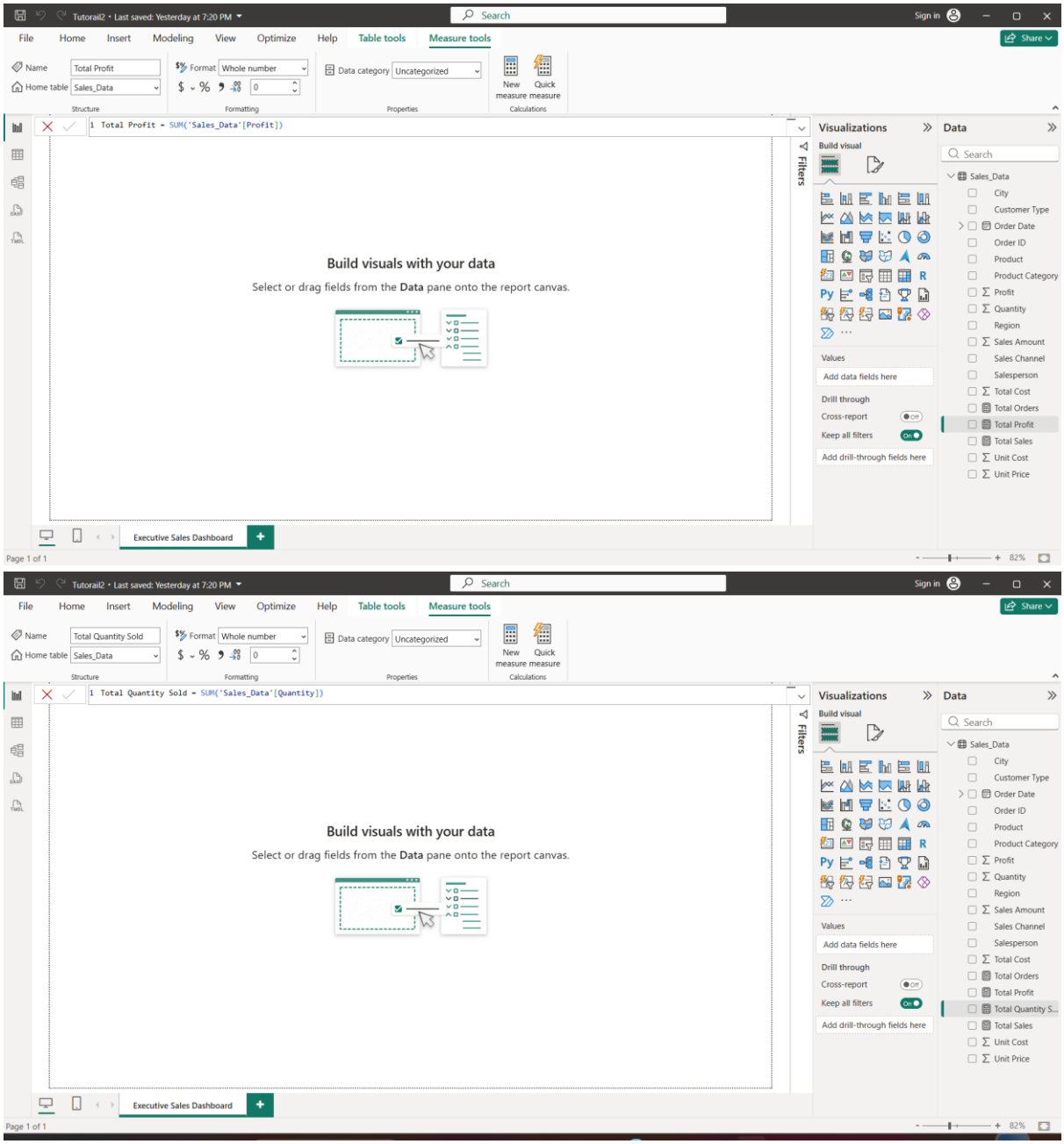

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Build visuals with your data
Select or drag fields from the Data pane onto the report canvas.
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Total Quantity Sold
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Executive Sales Dashboard
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Replaced Value6
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Added Custom1
Added Custom2
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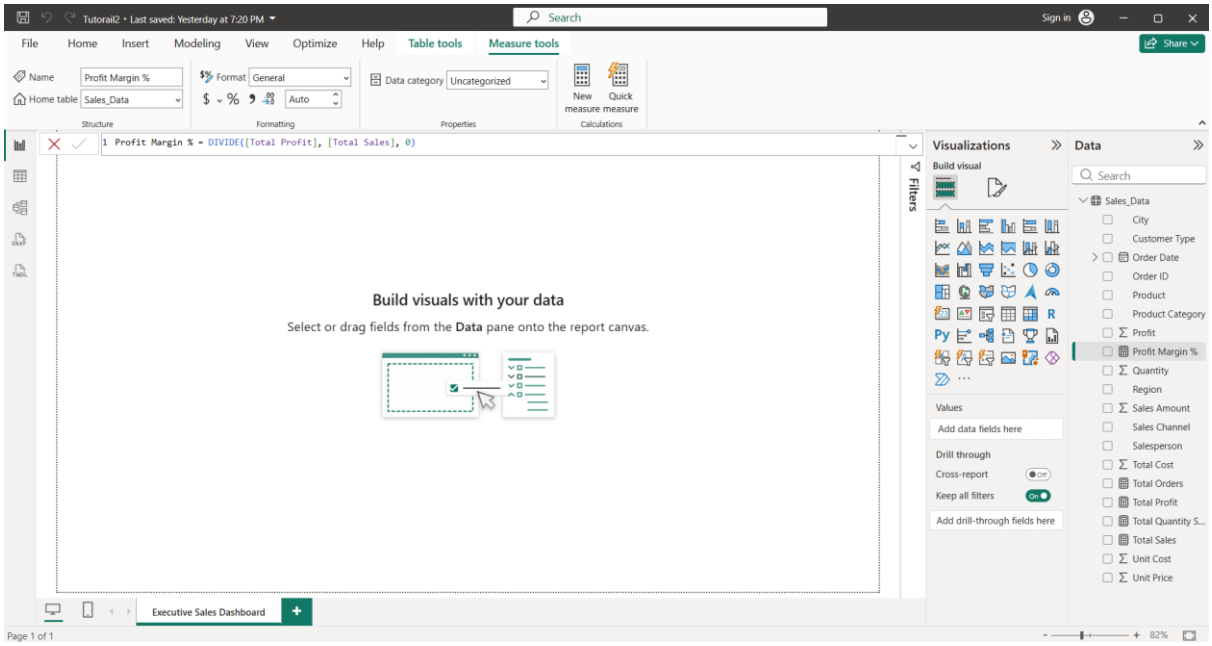
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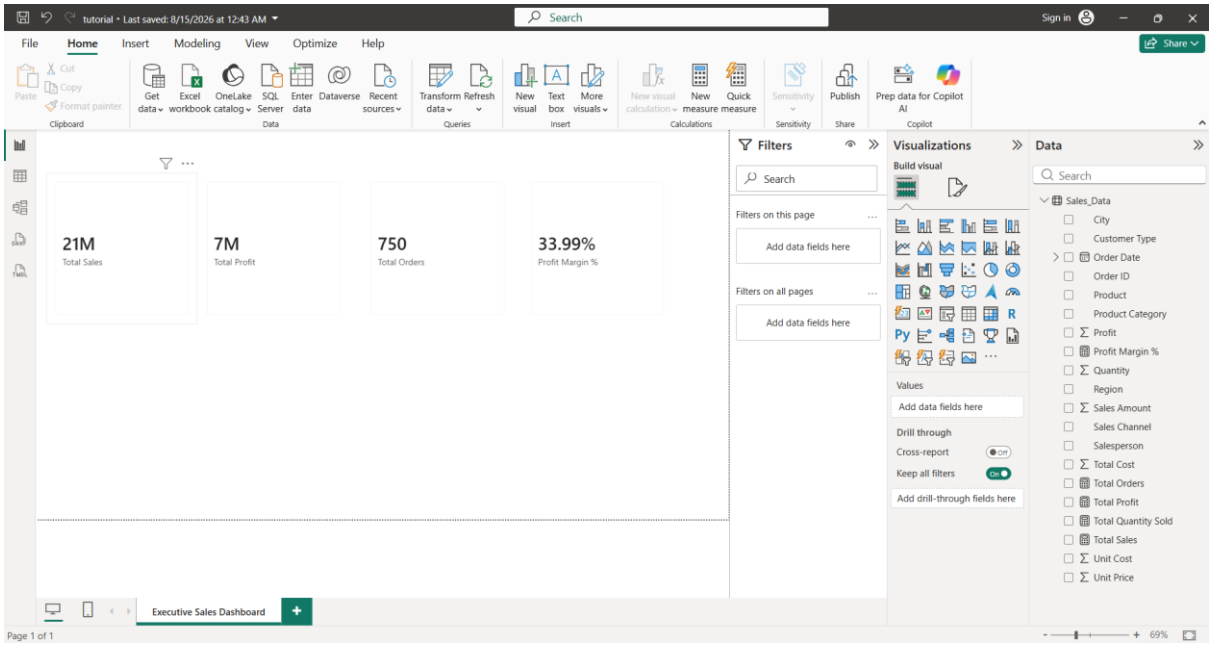


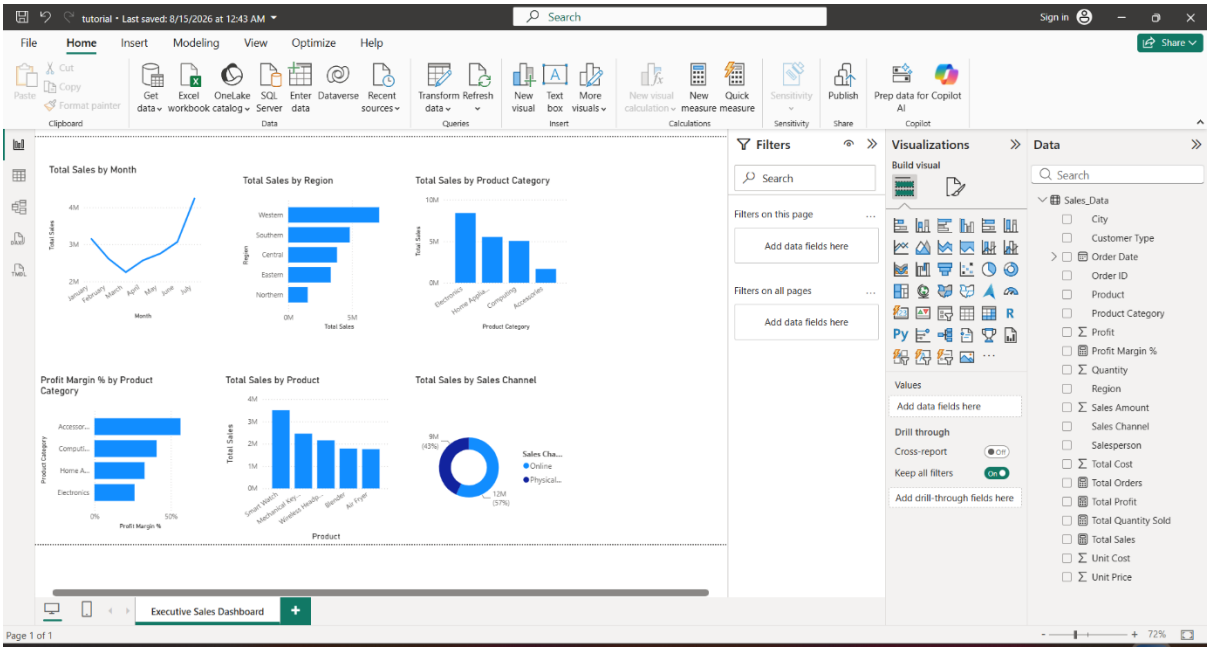

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Build visuals with your data
Select or drag fields from the Data pane onto the report canvas.
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Executive Sales Dashboard
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Format
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Uncategorized
New measure
Quick measure
Structure
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Calculations
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Build visuals with your data
Select or drag fields from the Data pane onto the report canvas.
Visualizations
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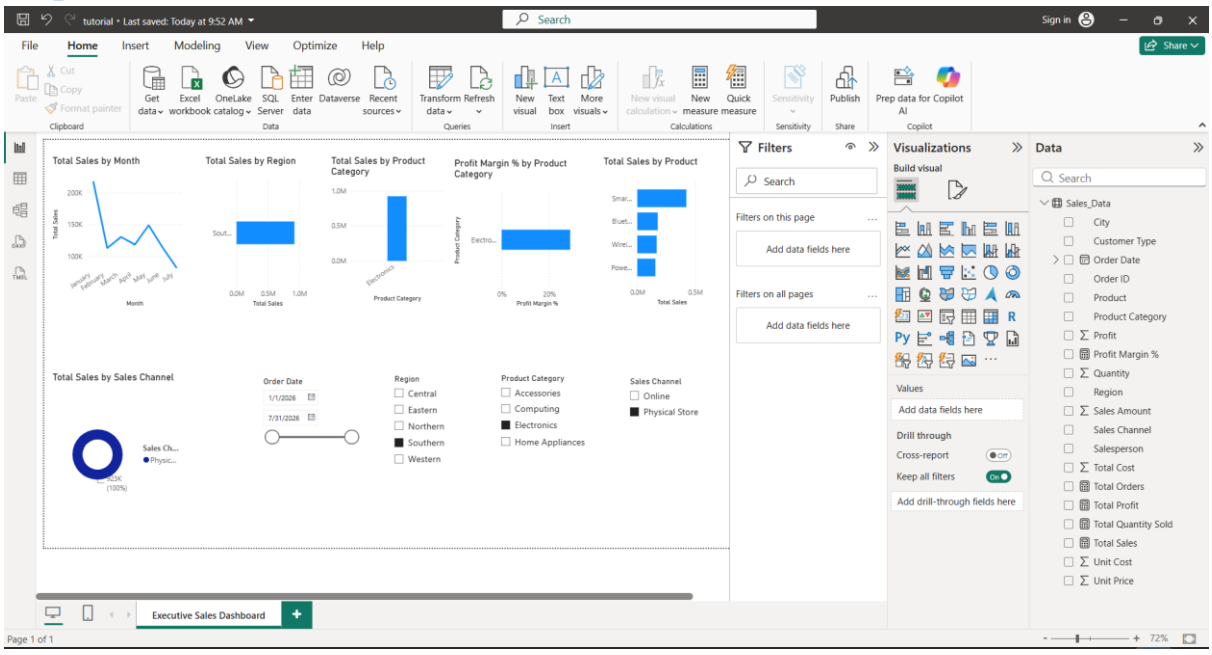


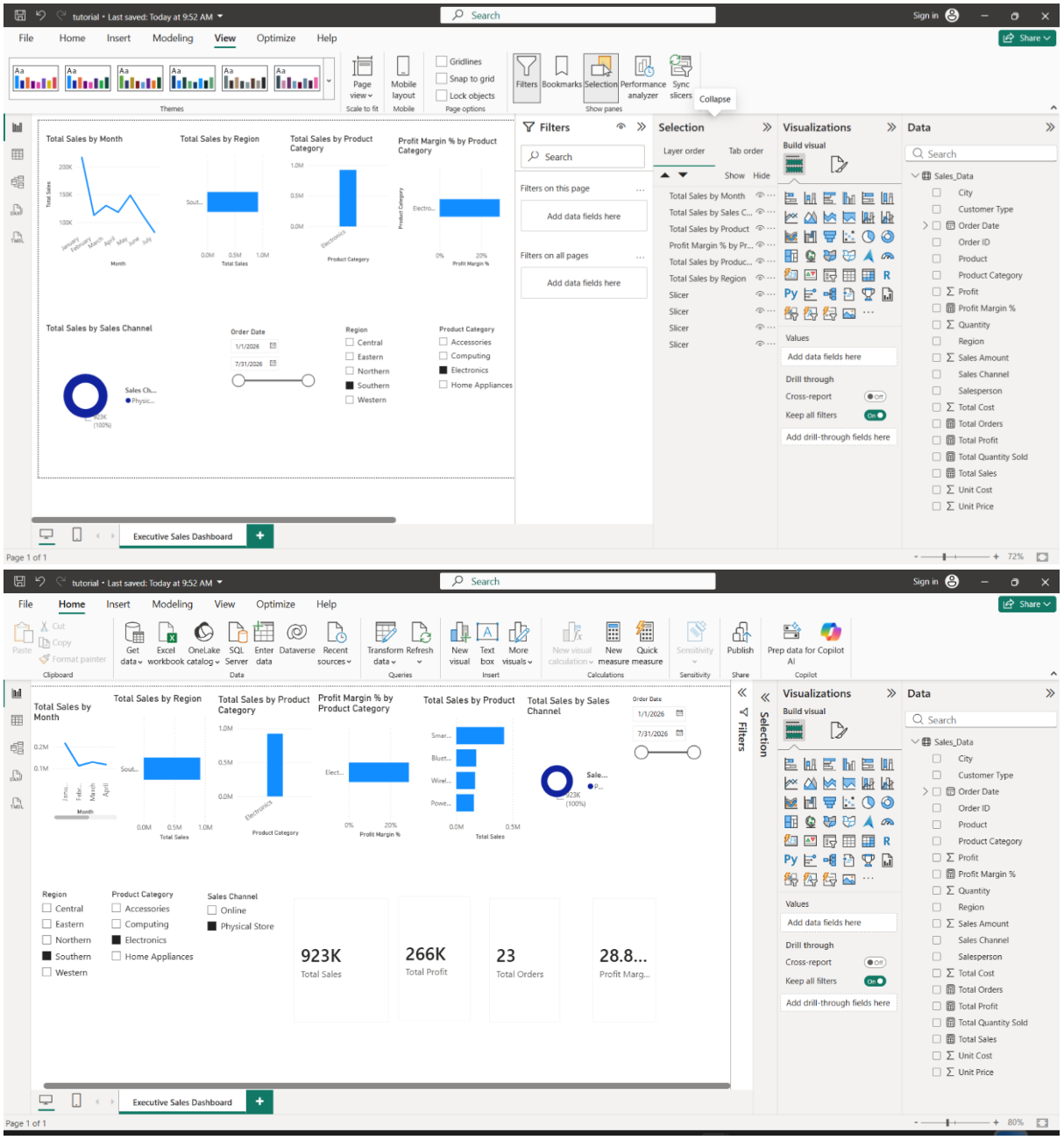
Step 7 - Build the Dashboard



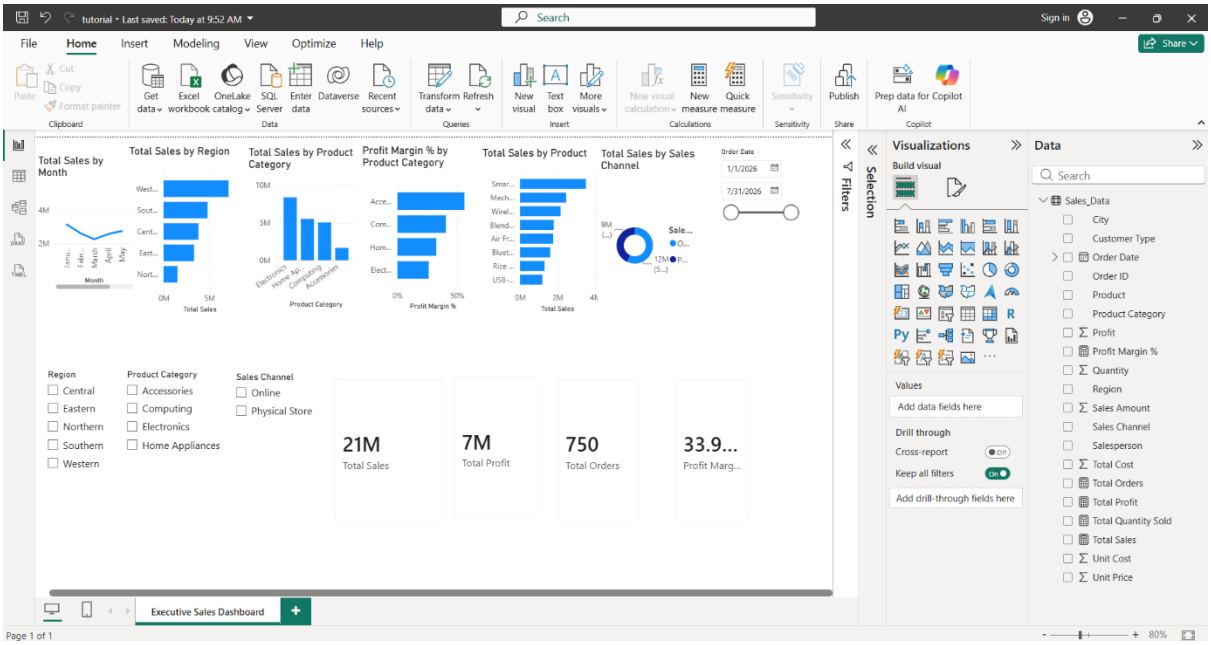


Step 8 - Add Slicers



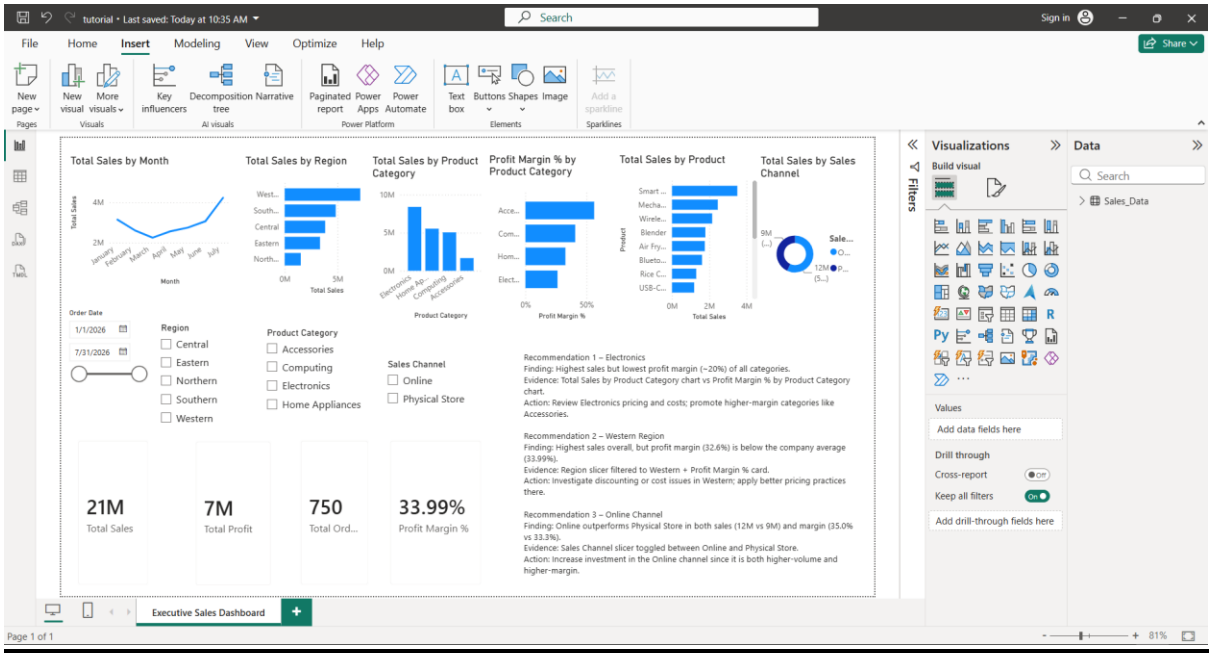

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Total Sales by Month
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Total Sales by Product Category
Profit Margin % by Product Category
Total Sales by Sales Channel
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Eastern
Northern
Southern
Western
Product Category
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Computing
Electronics
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OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for Copilot AI
Copilot
Total Sales by Month
Total Sales by Region
Total Sales by Product Category
Profit Margin % by Product Category
Total Sales by Product
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Sales Amount
Sales Channel
Salesperson
Total Cost
Total Orders
Total Profit
Total Quantity Sold
Total Sales
Unit Cost
Unit Price
Executive Sales Dashboard
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Final Dashboard

Executive Sales Dashboard, showing 4 KPI cards, 6 required charts, and 4 interactive slicers (Order Date, Region, Product Category, Sales Channel), with full unfiltered totals.



Step 9 – Business Questions & Answers

Q1. Which region has the highest Total Sales?

The Western region has the highest Total Sales, clearly the longest bar in the "Total Sales by Region" chart.

Q2. Does the highest-sales region also have the highest Profit Margin %?

No. When filtered to Western, the Profit Margin % is 32.6%, which is slightly below the company-wide average of 33.99%. The top-selling region is not the most profitable one relative to its sales.

Q3. Which Product Category generates the highest revenue?

Electronics generates the highest revenue — the tallest bar in the "Total Sales by Product Category" chart.

Q4. Which category has high sales but weaker profitability?

Electronics. Despite having the highest total sales, it has the lowest profit margin (about 20%) of all product categories, while Accessories, which sells far less, has the strongest margin (around 50%).

Q5. Is Online or Physical Store more profitable?

Online is more profitable. Online sales total 12M with a profit margin of about 35.0%, while Physical Store totals 9M in sales with a margin of about 33.3%. Online also drives the majority of total sales (about 57%).

Q6. Which month has the strongest performance?

January has the strongest performance, with 668,300 in Total Sales — the highest of all months. Sales trend downward after January, with some fluctuation in between (Feb 455.2K, Mar 614K, Apr 379.9K, May 574.7K, Jun 590.2K, Jul 423.4K).

Q7. Which three products should management prioritize?

Smart Watch, Mechanical Keyboard, and Wireless Headphones — the top three products by Total Sales in the "Total Sales by Product" chart.

Q8. Identify one business problem that could be hidden by looking only at Total Sales.

Total Sales alone hides profitability. Electronics and the Western region both look like top performers by revenue, but both actually underperform on profit margin. A dashboard built around Total Sales alone would push management toward over-investing in high-revenue but lower-margin areas, while overlooking smaller but more efficient ones like Accessories or the Online channel.

Step 10 – Management Recommendations

Recommendation 1 – Electronics

Finding: Highest sales but the lowest profit margin (~20%) of all product categories.

Evidence: “Total Sales by Product Category” chart compared with “Profit Margin % by Product Category” chart.

Action: Review Electronics pricing and cost structure; shift marketing emphasis toward higher-margin categories such as Accessories.

Recommendation 2 – Western Region

Finding: Highest sales overall, but profit margin (32.6%) is below the company average (33.99%).

Evidence: Region slicer filtered to Western, compared against the Profit Margin % card.

Action: Investigate discounting or cost issues specific to the Western region; apply pricing practices from higher-margin regions there.

Recommendation 3 – Online Channel

Finding: Online outperforms Physical Store in both total sales (12M vs 9M) and profit margin (35.0% vs 33.3%).

Evidence: Sales Channel slicer toggled between Online and Physical Store, comparing Total Sales and Profit Margin % cards.

Action: Increase investment in the Online channel (marketing, website experience, delivery speed) since it is both the higher-volume and higher-margin channel.